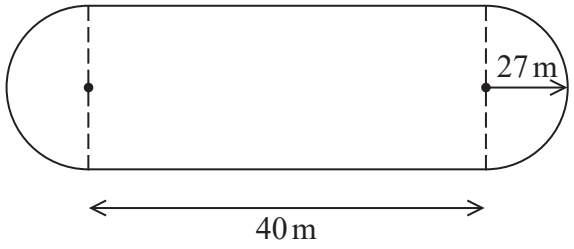
5 The diagram shows a cycle track.



The track has two straight sides each of length 40 m.
Each end of the track is a semicircle of radius 27 m.

The diameter of each wheel of Ian's bike is 590 mm.
Ian is going to ride his bike around the track once.

Calculate how many complete revolutions each wheel of his bike will make.

Two semicircles make one full circle of radius27m.

Curved distance = $2\pi \times 27 = 169.646003...$m.

1 mark
Combine curved ends

One lap = $2 \times 40 + 169.646003... = 249.646003...$m.

1 mark
Whole lap

Wheel diameter = 590mm = 0.590m.

Wheel circumference = $\pi \times 0.590 = 1.853539666...$m.

1 mark
Wheel circumference

Revolutions = $249.646003... \div 1.853539666...$

1 mark
Distance ÷ circumference

= 134.686086...; therefore134 complete revolutions.

1 mark
134 complete turns

(Total for Question 5 is 5 marks)